

ADVANCED

Stage 6

Functions (Adv), F2 Graphing (Adv)

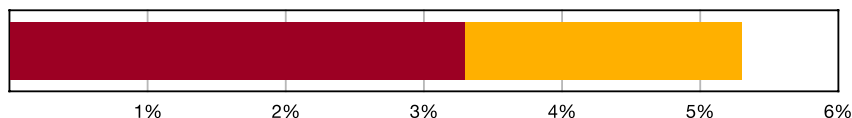
Transformations (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 81 minutes (based on allocation of 1.5 minutes per mark)



F2 Graphing



*SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.

HISTORICAL CONTRIBUTION

- F2 Graphing Techniques* has contributed an average of 5.3% per Advanced exam since the new syllabus was introduced in 2020.
- We have split the topic into 2 categories for analysis purposes: 1-*Transformations* (3.3%) and 2-*Non-Calculus Graphing* (2.0%).
- This analysis looks at *Transformations*.

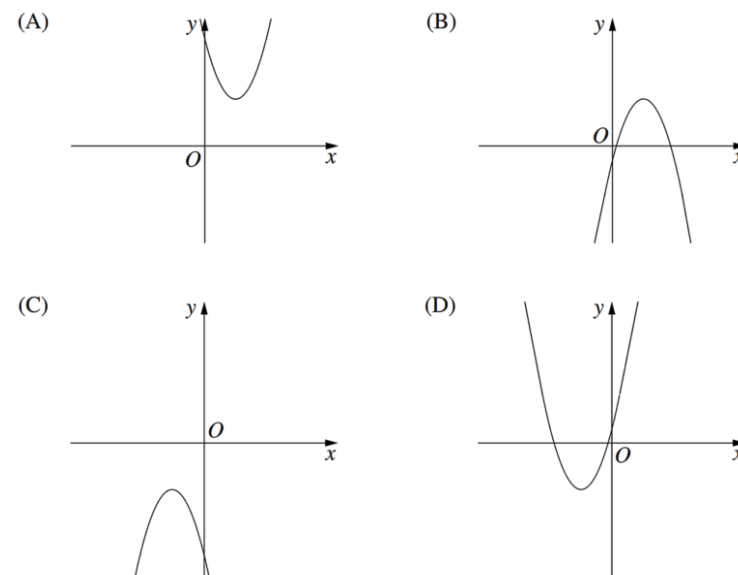
HSC ANALYSIS - What to expect and common pitfalls

- Transformations* (3.3%) represents new syllabus content that explicitly looks at translations and dilations of several function types, including the introduction of the aforementioned terminology.
- Absolute value* graph transformations were the subject of a 5-mark question in 2023 Adv 27. A "must revise" question in which particular attention should be paid to part (b) that produced a 23% mean mark.
- The 2022 Adv exam required students to calculate translations and dilations in three separate steps, producing a mean mark of 51%. This question is on the back of 2021 Q21 which combined vertical and horizontal dilations and similarly caused problems with a 48% state mean mark. Revision attention here goes without saying.
- The NESA sample HSC exam, released in March 2020, has been instructive in developing this challenging database area. Pay careful attention to *F2 EQ-Bank* questions.
- We note that Trig transformations, which we regard as an extremely important transformation sub-topic, are covered separately under *T3 Trig Graphs*.

Questions

1. Functions, 2ADV F2 2016 HSC 3 MC

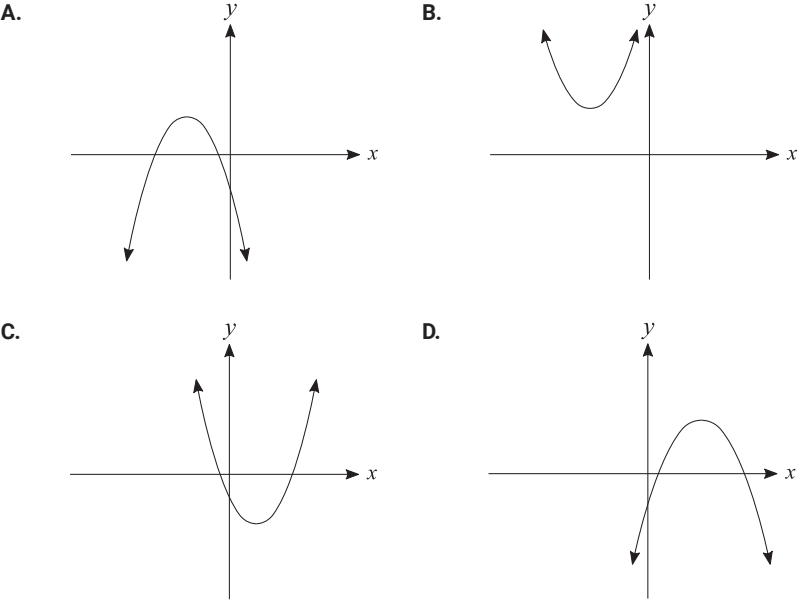
Which diagram best shows the graph of the parabola $y = 3 - (x - 2)^2$?



2. Functions, 2ADV F2 EQ-Bank 2 MC

Which diagram best shows the graph

$y = 1 - 2(x + 1)^2$



3. Functions, 2ADV F2 2020 HSC 2 MC

The function $f(x) = x^3$ is transformed to $g(x) = (x - 2)^3 + 5$ by a horizontal translation of 2 units followed by a vertical translation of 5 units.

Which row of the table shows the directions of the translations?

	Horizontal translation of 2 units	Vertical translation of 5 units
A.	Left	Up
B.	Right	Up
C.	Left	Down
D.	Right	Down

4. Functions, 2ADV F2 SM-Bank 5 MC

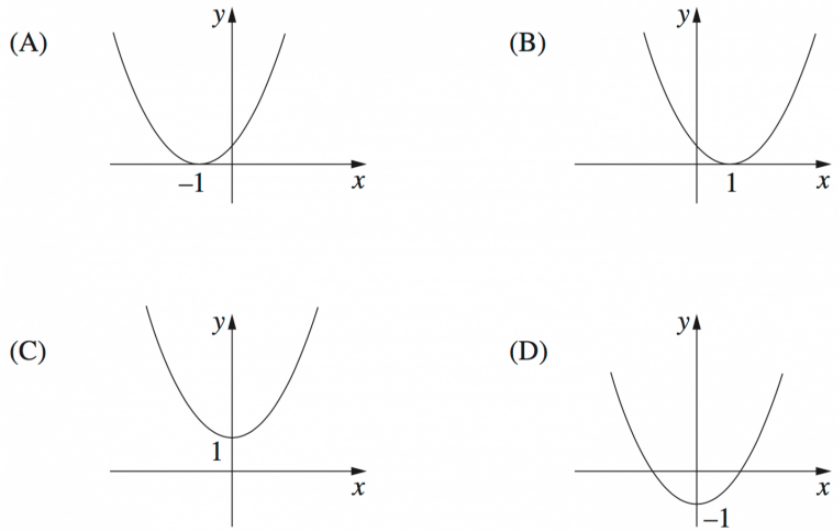
The point $P(4, -3)$ lies on the graph of a function $f(x)$. The graph of $f(x)$ is translated four units vertically up and then reflected in the y -axis.

The coordinates of the final image of P are

- A. $(-4, 1)$
- B. $(-4, 3)$
- C. $(0, -3)$
- D. $(4, -6)$

5. Functions, 2ADV F2 2014 HSC 2 MC

Which graph best represents $y = (x - 1)^2$?



6. Functions, 2ADV F2 2024 HSC 4 MC

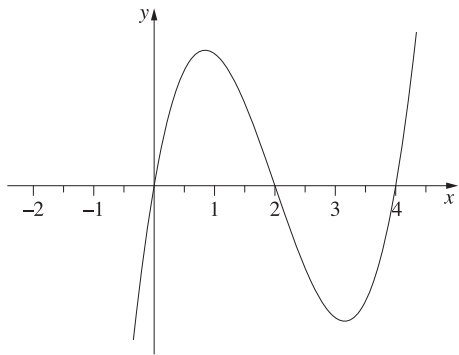
The parabola $y = (x - 3)^2 - 2$ is reflected about the y -axis. This is then reflected about the x -axis.

What is the equation of the resulting parabola?

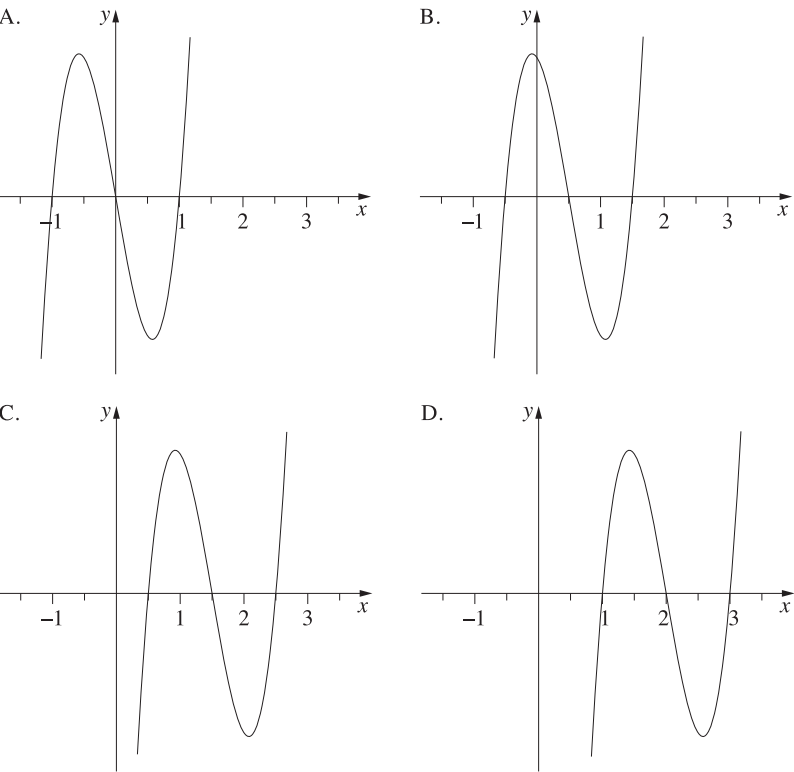
- A. $y = (x + 3)^2 + 2$
- B. $y = (x - 3)^2 + 2$
- C. $y = -(x + 3)^2 + 2$
- D. $y = -(x - 3)^2 + 2$

7. Functions, 2ADV F2 2024 HSC 7 MC

The diagram shows the graph $y = f(x)$.



Which of the following best represents the graph $y = f(2x - 1)$?



8. Functions, 2ADV F2 SM-Bank 6 MC

The graph of a function $f(x)$ is obtained from the graph of the function $g(x) = \sqrt{2x - 5}$ by a reflection in the x -axis followed by a dilation from the y -axis by a factor of $\frac{1}{2}$.

Which one of the following is the function $f(x)$?

- A. $f(x) = \sqrt{5 - 4x}$
- B. $f(x) = -\sqrt{x - 5}$
- C. $f(x) = \sqrt{x + 5}$
- D. $f(x) = -\sqrt{4x - 5}$

9. Functions, 2ADV F2 SM-Bank 4 MC

The graph of the function $f(x) = 3x^{\frac{5}{2}}$ is reflected in the x -axis and then translated 3 units to the right and 4 units down.

The equation of the new graph is

- A. $y = 3(x - 3)^{\frac{5}{2}} + 4$
- B. $y = -3(x - 3)^{\frac{5}{2}} - 4$
- C. $y = -3(x + 3)^{\frac{5}{2}} - 1$
- D. $y = -3(x - 4)^{\frac{5}{2}} + 3$

10. Functions, 2ADV F2 SM-Bank 7 MC

The point $A(3, 2)$ lies on the graph of the function $f(x)$. A transformation maps the graph of $f(x)$ to the graph of $g(x)$,

where $g(x) = \frac{1}{2}f(x - 1)$. The same transformation maps the point A to the point P .

The coordinates of the point P are

- A. $(2, 1)$
- B. $(2, 4)$
- C. $(4, 1)$
- D. $(4, 2)$

11. Functions, 2ADV F2 SM-Bank 8 MC

The transformation that maps the graph of $y = \sqrt{8x^3 + 1}$ onto the graph of $y = \sqrt{x^3 + 1}$ is a

- A. dilation by a factor of **2** from the y -axis.
- B. dilation by a factor of **2** from the x -axis.
- C. dilation by a factor of $\frac{1}{2}$ from the x -axis.
- D. dilation by a factor of $\frac{1}{2}$ from the y -axis.

12. Functions, 2ADV F2 2006 HSC 1c

Sketch the graph of $y = |x + 4|$. (2 marks)

13. Functions, 2ADV F2 SM-Bank 1

- i. Draw the graph $y = \ln x$. (1 mark)
- ii. Explain how the above graph can be transformed to produce the graph

$y = 3\ln(x + 2)$

and sketch the graph, clearly identifying all intercepts. (3 marks)

14. Functions, 2ADV F1 SM-Bank 35

- i. Sketch the function $y = f(x)$ where $f(x) = (x - 1)^3$ on a number plane, labelling all intercepts. (1 mark)
- ii. On the same graph, sketch $y = -f(-x)$. Label all intercepts. (2 marks)

15. Functions, 2ADV F2 EQ-Bank 16

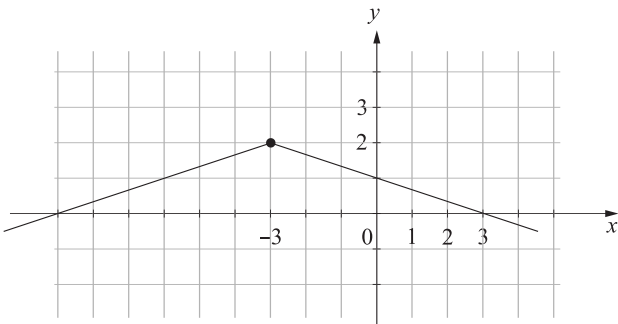
$y = -\frac{(x + 2)^4}{3}$ has been produced by three successive transformations: a translation, a dilation and then a reflection.

- i. Describe each transformation and state the equation of the graph after each transformation. (2 marks)
- ii. Sketch the graph. (1 mark)

16. Functions, 2ADV F2 EQ-Bank 1

The function $f(x) = |x|$ is transformed and the equation of the new function is $y = kf(x + b) + c$.

The graph of the new function is shown below.



What are the values of k , b and c . (2 marks)

17. Functions, 2ADV F2 EQ-Bank 14

List a set of transformations that, when applied in order, would transform $y = x^2$ to the graph with equation $y = 1 - 6x - x^2$. (3 marks)

18. Functions, 2ADV F2 SM-Bank 16

Let $f(x) = x^2 - 4$

Let the graph of $g(x)$ be a transformation of the graph of $f(x)$ where the transformations have been applied in the following order:

- dilation by a factor of $\frac{1}{2}$ from the vertical axis (parallel to the horizontal axis)
- translation by two units to the right (in the direction of the positive horizontal axis)

Find $g(x)$ and the coordinates of the horizontal axis intercepts of the graph of $g(x)$. (3 marks)

19. Functions, 2ADV F2 SM-Bank 2

Sketch the graph $y = \log_2(x - 3)$.

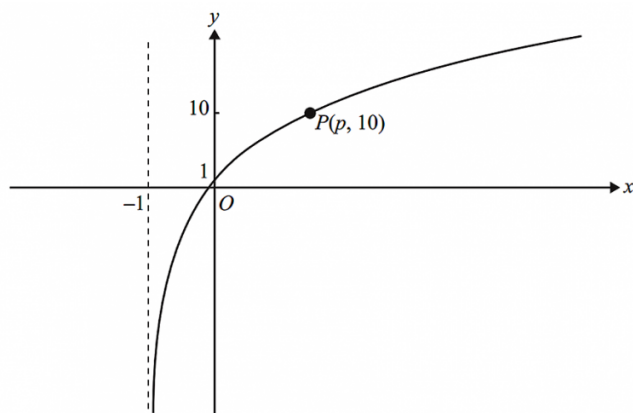
Show all asymptotes and state its domain and range. (3 marks)

20. Functions, 2ADV F2 SM-Bank 3

The diagram below shows part of the graph of the function with rule

$$f(x) = k \log_e(x + a) + c, \text{ where } k, a \text{ and } c \text{ are real constants.}$$

- The graph has a vertical asymptote with equation $x = -1$.
- The graph has a y -axis intercept at 1.
- The point P on the graph has coordinates $(p, 10)$, where p is another real constant.



- State the value of a . (1 mark)
- Find the value of c . (1 mark)
- Show that $k = \frac{9}{\log_e(p + 1)}$. (2 marks)

21. Functions, 2ADV F2 EQ-Bank 13

The curve $y = kx^2 + c$ is subject to the following transformations

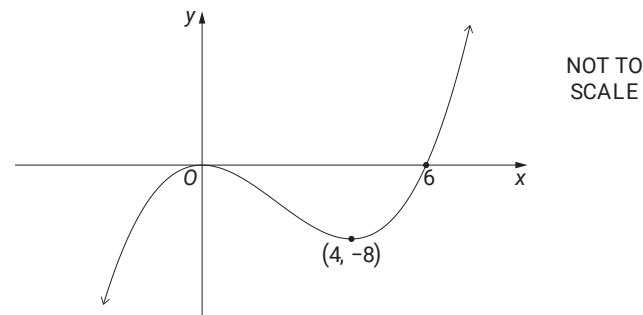
- Translated 2 units in the positive x -direction
- Dilated in the positive y -direction by a factor of 4
- Reflected in the y -axis

The final equation of the curve is $y = 8x^2 + 32x - 8$.

- Find the equation of the graph after the dilation. (1 mark)
- Find the values of k and c . (2 marks)

22. Functions, 2ADV F2 2021 HSC 21

Consider the graph of $y = f(x)$ as shown.



Sketch the graph of $y = 4f(2x)$ showing the x -intercepts and the coordinates of the turning points. (2 marks)

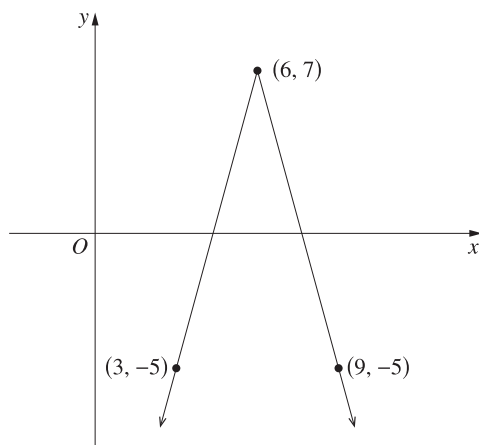
23. Functions, 2ADV F2 2022 HSC 19

The graph of the function $f(x) = x^2$ is translated m units to the right, dilated vertically by a scale factor of k and then translated 5 units down. The equation of the transformed function is $g(x) = 3x^2 - 12x + 7$.

Find the values of m and k . (3 marks)

24. Functions, 2ADV F2 2023 HSC 27

The graph of $y = f(x)$, where $f(x) = a|x - b| + c$, passes through the points $(3, -5)$, $(6, 7)$ and $(9, -5)$ as shown in the diagram.



- Find the values of a , b and c . (3 marks)
- The line $y = mx$ cuts the graph of $y = f(x)$ in two distinct places.
Find all possible values of m . (2 marks)

25. Functions, 2ADV F2 2013 HSC 15c

- Sketch the graph $y = |2x - 3|$. (1 mark)
- Using the graph from part (i), or otherwise, find all values of m for which the equation $|2x - 3| = mx + 1$ has exactly one solution. (2 marks)

Worked Solutions

1. Functions, 2ADV F2 2016 HSC 3 MC

$$y = 3 - (x - 2)^2$$

Curve is concave down and passes through $(2, 3)$

$\Rightarrow B$

2. Functions, 2ADV F2 EQ-Bank 2 MC

Transforming $y = x^2$:

Translate 1 unit left $\Rightarrow y = (x + 1)^2$

Dilate from x -axis by a factor of 2 $\Rightarrow y = 2(x + 1)^2$

Reflect in x -axis $\Rightarrow y = -2(x + 1)^2$

Translate 1 unit up $\Rightarrow y = 1 - 2(x + 1)^2$

$\therefore$ Transformations describe graph A .

$\Rightarrow A$

3. Functions, 2ADV F2 2020 HSC 2 MC

Horizontal translation: 2 units to the right

$$x^3 \rightarrow (x - 2)^3$$

Vertical translation: 5 units up

$$(x - 2)^3 \rightarrow (x - 2)^3 + 5$$

$\Rightarrow B$

4. Functions, 2ADV F2 SM-Bank 5 MC

1st transformation:

$$P(4, -3) \rightarrow (4, 1)$$

2nd transformation:

$$(4, 1) \rightarrow (-4, 1)$$

$\Rightarrow A$

5. Functions, 2ADV F2 2014 HSC 2 MC

$$y = (x - 1)^2 \rightarrow \text{vertex } (1, 0)$$

$\Rightarrow B$

6. Functions, 2ADV F2 2024 HSC 4 MC

$y = (x - 3)^2 - 2$

Reflect in the y -axis ($f(-x)$):

$y = (-x - 3)^2 - 2$

Reflect in the x -axis ($-f(-x)$):

$y = -[(-x - 3)^2 - 2]$
 $= -(x + 3)^2 + 2$

$\Rightarrow C$

7. Functions, 2ADV F2 2024 HSC 7 MC

At $x = 0$:

$f(2x - 1) = f(-1) \Rightarrow$ Eliminate A and B .

Consider the transformations of $f(x) \rightarrow f(2x - 1)$

$\rightarrow$ Shift $f(x)$ 1 unit to the right.

$\rightarrow$ Dilate $f(x)$ by a factor of $\frac{1}{2}$ from the y -axis.

$\Rightarrow C$

8. Functions, 2ADV F2 SM-Bank 6 MC

Let $y = \sqrt{2x - 5}$

1st transformation:

$y = -\sqrt{2x - 5}$

2nd transformation:

Dilate from y -axis by a factor of $\frac{1}{2}$

$\Rightarrow$ Swap $x \rightarrow 2x$

$y = -\sqrt{2(2x) - 5}$
 $= -\sqrt{4x - 5}$

$\therefore f(x) = -\sqrt{4x - 5}$

$\Rightarrow D$

COMMENT: Using "swap" terminology for dilations from the y -axis is simpler and more intelligible for students in our view.

9. Functions, 2ADV F2 SM-Bank 4 MC

Let $y = 3x^{\frac{5}{2}}$

Reflect in the x -axis:

$y = -3x^{\frac{5}{2}}$

Translate 3 units to the right:

$y = -3(x - 3)^{\frac{5}{2}}$

Translate 4 units down:

$y = -3(x - 3)^{\frac{5}{2}} - 4$

$\Rightarrow B$

10. Functions, 2ADV F2 SM-Bank 7 MC

$g(x) = \frac{1}{2}f(x-1), A(3, 2)$

Dilate by a factor of $\frac{1}{2}$ from x -axis:

$A(3, 2) \rightarrow A'(3, 1)$

Translate 1 unit to right:

$A'(3, 1) \rightarrow P(4, 1)$

$\Rightarrow C$

11. Functions, 2ADV F2 SM-Bank 8 MC

Let $f(x) = \sqrt{8x^3 + 1}$

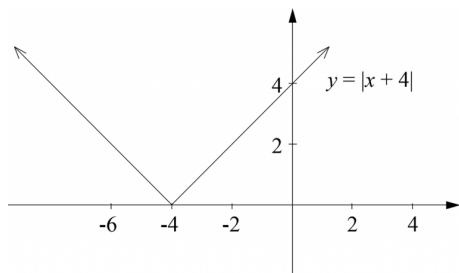
$f\left(\frac{1}{2}x\right) = \sqrt{8\left(\frac{1}{2}x\right)^3 + 1}$
 $= \sqrt{x^3 + 1}$

$\therefore$ Transformation correct when x is swapped for $\frac{x}{2}$

i.e. graph is dilated by factor of 2 from y -axis

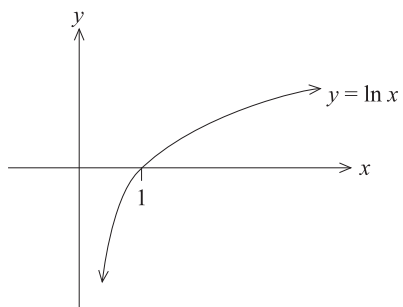
$\Rightarrow A$

12. Functions, 2ADV F2 2006 HSC 1c



13. Functions, 2ADV F2 SM-Bank 1

i.

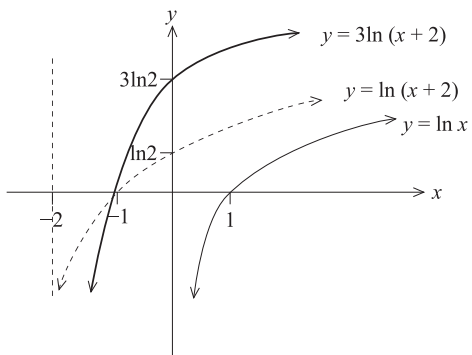


ii. Transforming $y = \ln x \Rightarrow y = \ln(x + 2)$

$y = \ln x \Rightarrow$ shift 2 units to left.

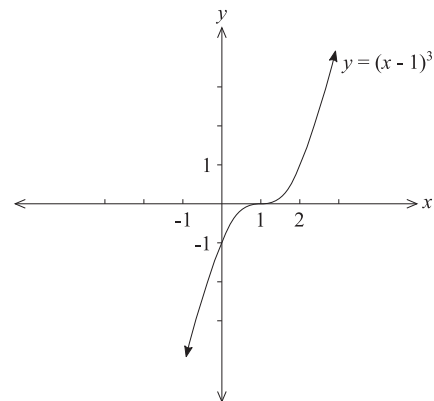
Transforming $y = \ln(x + 2)$ to $y = 3\ln(x + 2)$

$\Rightarrow$ increase each y value by a factor of 3



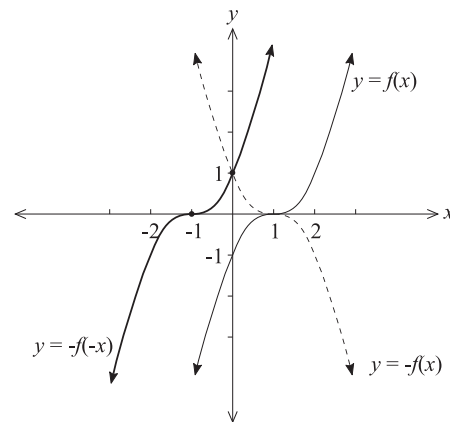
14. Functions, 2ADV F1 SM-Bank 35

i. $y = (x-1)^3 \Rightarrow y = x^3$ shifted 1 unit to the right.



ii. $y = -f(x) \Rightarrow$ reflect $y = (x-1)^3$ in x -axis.

$y = -f(-x) \Rightarrow$ reflect $y = -f(x)$ in y -axis.



15. Functions, 2ADV F2 EQ-Bank 16

i. Transformation 1:

Translate $y = x^4$ 2 units to the left.

$$y = x^4 \Rightarrow y = (x + 2)^4$$

Transformation 2:

Dilate $y = (x + 2)^4$ by a factor of $\frac{1}{3}$ from the x -axis

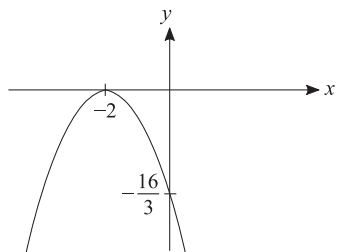
$$y = (x + 2)^4 \Rightarrow y = \frac{(x + 2)^4}{3}$$

Transformation 3:

Reflect $y = \frac{(x + 2)^4}{3}$ in the x -axis.

$$y = \frac{(x + 2)^4}{3} \Rightarrow y = -\frac{(x + 2)^4}{3}$$

ii.



16. Functions, 2ADV F2 EQ-Bank 1

$$y = |x|$$

Translate 3 units left $\Rightarrow y = |x + 3|$

Reflect in the x -axis $\Rightarrow y = -|x + 3|$

Dilate by $\frac{1}{3}$ from the x -axis

$$\Rightarrow \text{Multiply by } \frac{1}{3} \Rightarrow y = -\frac{1}{3}|x + 3|$$

Translate 2 units up $\Rightarrow y = -\frac{1}{3}|x + 3| + 2$

$$\therefore k = -\frac{1}{3}, b = 3, c = 2$$

17. Functions, 2ADV F2 EQ-Bank 14

$$y = x^2$$

Transformation 1:

Translate 3 units in negative x -direction

$$y = (x + 3)^2$$

$$y = x^2 + 6x + 9$$

Transformation 2:

Translate 10 units in negative y -direction

$$y = x^2 + 6x - 1$$

Transformation 3:

Reflect in the x -axis

$$y = -(x^2 + 6x - 1) \\ = 1 - 6x - x^2$$

18. Functions, 2ADV F2 SM-Bank 16

1st transformation

Dilation by a factor of $\frac{1}{2}$ from the y -axis:

$$x^2 - 4 \Rightarrow \left(\frac{x}{\frac{1}{2}}\right)^2 - 4 = 4x^2 - 4$$

2nd transformation

Translation by 2 units to the right:

$$4x^2 - 4 \Rightarrow g(x) = 4(x - 2)^2 - 4$$

x -axis intercept of $g(x)$:

$$4(x - 2)^2 - 4 = 0$$

$$(x - 2)^2 = 1$$

$$x - 2 = \pm 1$$

$$x - 2 = 1 \Rightarrow x = 3$$

$$x - 2 = -1 \Rightarrow x = 1$$

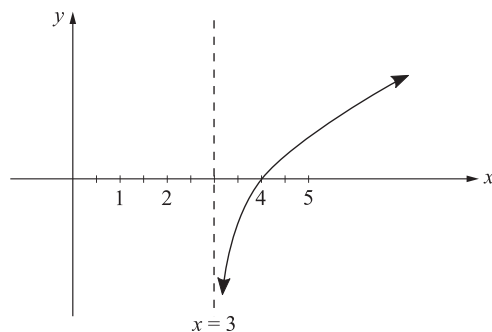
$\therefore$ Horizontal axis intercepts occur at $(1, 0)$ and $(3, 0)$.

19. Functions, 2ADV F2 SM-Bank 2

Asymptote when $x = 3$

Domain $\{x: x > 3\}$

Range: all y



20. Functions, 2ADV F2 SM-Bank 3

i. Vertical Asymptote:

$$x = -1$$

$$\therefore a = 1$$

ii. Solve $f(0) = 1$ for c ,

$$c = 1$$

iii. $f(x) = k \log_e(x + 1) + 1$

Since $f(p) = 10$,

$$k \log_e(p + 1) + 1 = 10$$

$$k \log_e(p + 1) = 9$$

$$\therefore k = \frac{9}{\log_e(p + 1)} \dots \text{as required}$$

21. Functions, 2ADV F2 EQ-Bank 13

i. $y = kx^2 + c$

Translate 2 units in positive x -direction.

$$y = kx^2 + c \Rightarrow y = k(x-2)^2 + c$$

Dilate in the positive y -direction by a factor of 4.

$$y = k(x-2)^2 + c \Rightarrow y = 4k(x-2)^2 + 4c$$

ii. $y = 4k(x^2 - 4x + 4) + 4c$

$$= 4kx^2 - 16kx + 16k + 4c$$

Reflect in the y -axis.

$$\Rightarrow \text{Swap: } x \rightarrow -x$$

$$y = 4k(-x)^2 - 16k(-x) + 16k + 4c$$

$$= 4kx^2 + 16kx + 16k + 4c$$

Equating co-efficients:

$$4k = 8$$

$$\therefore k = 2$$

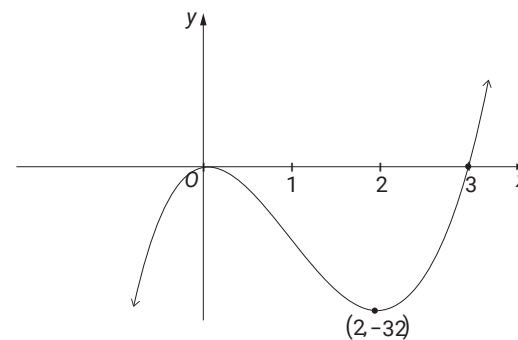
$$16k + 4c = -8$$

$$4c = -40$$

$$\therefore c = -10$$

COMMENT: Using "swap" terminology for reflections in the y -axis is simpler and more intelligible for students in our view.

22. Functions, 2ADV F2 2021 HSC 21



♦ Mean mark 48%.

23. Functions, 2ADV F2 2022 HSC 19

Horizontal translation m units to the right:

$$x^2 \rightarrow (x - m)^2$$

Dilated vertically by scale factor k :

$$(x - m)^2 \rightarrow k(x - m)^2$$

Vertical translation 5 units down:

$$k(x - m)^2 \rightarrow k(x - m)^2 - 5$$

$$\begin{aligned} y &= k(x - m)^2 - 5 \\ &= k(x^2 - 2mx + m^2) - 5 \\ &= kx^2 - 2kmx + (km^2 - 5) \end{aligned}$$

$$\begin{aligned} \therefore k &= 3 \\ -2km &= -12 \\ \therefore m &= 2 \end{aligned}$$

◆ Mean mark 51%.

24. Functions, 2ADV F2 2023 HSC 27

a. Consider the transformation of $y = -|x|$

Translate 6 units to the right

$$\begin{aligned} y &= -|x| \rightarrow y = -|x - 6| \\ \therefore b &= 6 \end{aligned}$$

Translate 7 units vertically up

$$\begin{aligned} y &= -|x - 6| \rightarrow y = -|x - 6| + 7 \\ \therefore c &= 7 \end{aligned}$$

$f(x) = a|x - 6| + 7$ passes through $(3, -5)$:

$$\begin{aligned} -5 &= a|3 - 6| + 7 \\ -5 &= 3a + 7 \\ 3a &= -12 \\ \therefore a &= -4 \end{aligned}$$

◆◆ Mean mark (b) 23%.

b. $y = mx$ passes through $(0, 0)$

One solution when $y = mx$ passes through $(0, 0)$ and $(6, 7)$

$$m = \frac{7 - 0}{6 - 0} = \frac{7}{6}$$

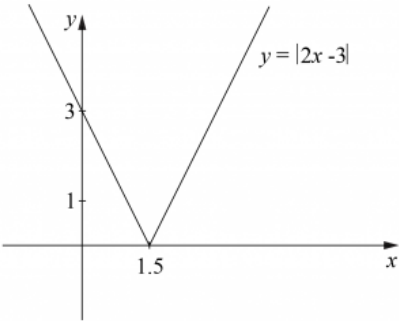
As graph gets flatter and turns negative $\Rightarrow$ 2 solutions

2 solutions continue until $y = mx$ is parallel to the line joining $(6, 7)$ to $(9, -5)$, where:

$$m = \frac{7 - (-5)}{6 - 9} = -\frac{12}{3} = -4$$

$$\therefore \text{2 solutions when } -4 < m < \frac{7}{6}$$

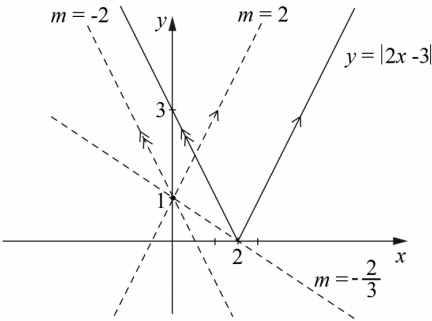
i.



♦ Mean mark 49%

MARKER'S COMMENT: Many students drew diagrams that were "too small", didn't use rulers or didn't use a consistent scale on the axes!

ii.



Line of intersection $y = mx + 1$ passes through $(0, 1)$

If it also passes through $(1.5, 0) \Rightarrow 1$ solution

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 - 0}{0 - \frac{3}{2}} \\ &= -\frac{2}{3} \end{aligned}$$

Gradients of $y = |2x - 3|$ are 2 or -2

Considering a line through $(0, 1)$:

If $m \geq 2$, only intersects once.

Similarly,

If $m < -2$, only intersects once.

$\therefore$ Only one solution when $m = -\frac{2}{3}$, $m \geq 2$ or $m < -2$

♦♦ Mean mark 25%.

COMMENT: Students need a clear graphical understanding of what they are finding to solve this very challenging, Band 6 question.